Aggregate-response pricing: the city's charges from observed zone aggregates alone, with the equilibrium response as a proximal map

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Theory notes, direction D

Abstract

The city of the competing-carrier model (`theory/main.tex`, D1; `theory/new_algorithm_C`) sets per-delivery and congestion charges for N carriers whose plans come from an NP-hard routing oracle. Dual implementation pricing (DIP) computes near-optimal charges but needs the carriers' own oracle: it solves their routing problems, which a real city cannot do. Here the city observes only the zone aggregates that the carriers' equilibrium produces, deliveries Q_z and the number n_z of active carriers, and nothing else. We prove that in the convexified game the equilibrium aggregate is the proximal map of the carriers' aggregate value function, $Q(u) = \nabla F^*(-u)$ (Theorem 3), so that the leader's problem is a strongly convex dual problem whose gradient is exactly the observed quantity $(u + \frac{m}{2}\bar{d})/\kappa - Q(u)$. The Pigou–Cournot update $u \leftarrow \kappa Q - \frac{m}{2}\bar{d}$ is therefore a gradient step, and it contracts at ratio $\rho = \max_z \kappa_z/(b_z + \kappa_z) < 1$ in the κ^{-1}-norm, for every instance, independently of the number of zones, carriers, customers and of the routing structure; the damped step $2/(2+\rho)$ contracts at ratio $\rho/(2+\rho)$ (Theorem 6). Its fixed point is the exact minimiser of the convexified social cost plus the lot-asymmetry term of DIP (Theorem 5). In the pure, lumpy game every fixed point is a unilateral social optimum up to lot asymmetry (Theorem 8), the iteration stays within an explicit neighbourhood of the convexified optimum (Theorem 9), and the response map identifies the convexified value function up to a constant, so the convexified optimum is the most that any observation-only scheme can target (Theorem 10). Belief imprecision of the carriers is absorbed: a Γ-maximin surcharge only shifts the fixed point (Proposition 12), and under E-admissibility of width δ the iteration implements a pure target exactly as soon as it is within the surcharge band of an implementing charge (Proposition 13). The limit of re-tangenting at every transaction is the realised-aggregate tariff $T_z(q, Q) = \kappa_z q(Q - q) - \frac{b_z}{2}q^2$, under which the social cost is the exact potential of the carriers' game (Theorem 14); it needs no target, no oracle and no iteration, every social optimum is an equilibrium under it, and the carriers' own best responses never raise the social cost. Section 7 says what is classical (trial-and-error marginal-cost pricing, adaptive incentive design, marginal-contribution tariffs) and what is new (the proximal identity and its instance-free rates, the lumpy fixed-point and identification results, the credal absorption of the leader's error). Experiments are in `results.md`.

1 Setting

We use the game and notation of `theory/new_algorithm_C/theory.tex` (DIP), restated for completeness.

Assumption 1 (City game, D1; `src/ccu/zmc_brl.py`)**.** Zones $z \in [Z]$, carriers $i \in [N]$. Carrier i has customer zones $C_i \subseteq [Z]$ with lot sizes $d_{iz} > 0$ for $z \in C_i$ ($d_{iz} = 0$ otherwise) and a finite plan set $\mathcal{P}_i$. A plan p serves a set $S(p) \subseteq C_i$ (all-or-nothing lots) with routes of routing cost $R_i(p) \geq 0$ (the BRL upper expectation). Deliveries are $q_{iz}(p) = d_{iz}\mathbf{1}[z \in S(p)]$ and vehicle-hours $\tau_{iz}(p) = r\, q_{iz}(p)$ for one ratio $r > 0$. Inverse demand is $\alpha_z - b_z Q_z$ and the congestion cost is $g_z X_z$ per vehicle-hour, with $Q = \sum_i q_i$ and $X = rQ$. The city posts a per-delivery-equivalent charge $u \in \mathbb{R}^Z$ (Proposition 1 of DIP: a uniform per-delivery charge plus non-negative zone congestion charges realise every u). Carrier i's cost at the pure profile σ is

$$J_i^u(p; \sigma_{-i}) = R_i(p) + \sum_z q_{iz}(p)\big(-\alpha_z + m_z(d_{iz} + Q_{-i,z}) + u_z\big), \qquad m_z = b_z + \kappa_z,\ \kappa_z = g_z r^2,$$

and the social cost (charges are transfers) is $\mathrm{SC}(\sigma) = \sum_i R_i(\sigma_i) + \sum_z \big[-\alpha_z Q_z + \frac{m_z^S}{2} Q_z^2\big]$ with $m_z^S = b_z + 2\kappa_z$.

Throughout, $q_{iz}^2 = d_{iz} q_{iz}$ because $q_{iz} \in \{0, d_{iz}\}$. The game has the exact potential (D1, Monderer–Shapley)

$$\Phi_u(\sigma) = \sum_i R_i(\sigma_i) + \sum_z \Big[-\alpha_z Q_z + \tfrac{m_z}{2}\big(Q_z^2 + \textstyle\sum_i q_{iz}^2\big) + u_z Q_z\Big]. \tag{1}$$

For $\bar{d} \in \mathbb{R}^Z$ put $L_{\bar{d}}(\sigma) = \sum_{i,z} \frac{m_z}{2}(d_{iz} - \bar{d}_z) q_{iz}$ (lot asymmetry) and $E_{\bar{d}} = \sum_{i,z} \frac{m_z}{2} |d_{iz} - \bar{d}_z| d_{iz}$.

Lemma 1 (Potential–welfare identity; DIP Lemma 2)**.** [PROVED] *For every $Q^\circ, \bar{d} \in \mathbb{R}^Z$ and $u = \kappa \odot Q^\circ - \frac{m}{2} \odot \bar{d} + v$,*

$$\Phi_u(\sigma) = \mathrm{SC}(\sigma) - \sum_z \tfrac{\kappa_z}{2}(Q_z - Q_z^\circ)^2 + L_{\bar{d}}(\sigma) + \langle v, Q\rangle + \sum_z \tfrac{\kappa_z}{2}(Q_z^\circ)^2. \tag{2}$$

Proof. $\Phi_u - \mathrm{SC} = \sum_z [\frac{m_z}{2} \sum_i d_{iz} q_{iz} - \frac{\kappa_z}{2} Q_z^2 + u_z Q_z]$ because $m_z^S - m_z = \kappa_z$ and $q_{iz}^2 = d_{iz} q_{iz}$. Insert u_z, complete the square $-\frac{\kappa_z}{2} Q_z^2 + \kappa_z Q_z^\circ Q_z = -\frac{\kappa_z}{2}(Q_z - Q_z^\circ)^2 + \frac{\kappa_z}{2}(Q_z^\circ)^2$ and write $\frac{m_z}{2} \sum_i d_{iz} q_{iz} - \frac{m_z}{2} \bar{d}_z Q_z = \sum_i \frac{m_z}{2}(d_{iz} - \bar{d}_z) q_{iz}$. □

What the city sees. The city knows the demand and congestion slopes b, g, r (hence κ, m) and nothing about R_i, $\mathcal{P}_i$ or d_{iz}. After posting u it observes the zone aggregates of the equilibrium the carriers settle in: Q_z and $n_z = \#\{i : q_{iz} > 0\}$.

The rule. With an estimate $\bar{d}$ of the mean lot (Section 6),

$$u_z^{t+1} = (1 - \theta_z) u_z^t + \theta_z \Big[\kappa_z Q_z^t - \tfrac{m_z}{2} \bar{d}_z\Big], \qquad \theta_z \in (0, 1]. \tag{3}$$

$\theta = 1$ is the plain Pigou–Cournot fixed-point iteration (DIP's tangent charge evaluated at the observed aggregate instead of the computed target).

2 The convexified game: the response is a proximal map

Let $\mathcal{X} = \prod_i \Delta(\mathcal{P}_i)$, $\bar{q}_i(x_i) = \mathbb{E}_{x_i} q_i$, $\bar{Q}(x) = \sum_i \bar{q}_i(x_i)$ and $\mathcal{Q} = \bar{Q}(\mathcal{X})$, a polytope in $\mathbb{R}^Z_{\geq 0}$. The convexified potential and social cost are (DIP, Section 3)

$$\overline{\Phi}_u(x) = \sum_i \mathbb{E}_{x_i}\Big[R_i + \sum_z q_{iz}\big(-\alpha_z + u_z + \tfrac{m_z}{2} d_{iz}\big)\Big] + \sum_z \tfrac{m_z}{2} \bar{Q}_z^2,$$

$$\overline{\mathrm{SC}}(x) = \sum_i \mathbb{E}_{x_i}\Big[R_i - \sum_z \alpha_z q_{iz}\Big] + \sum_z \tfrac{m_z^S}{2} \bar{Q}_z^2.$$

Their minimisers are the convexified equilibria and the convexified social optima. Fix a reference $\bar{d}$ and define, for $Q \in \mathcal{Q}$,

$$G(Q) = \min \Big\{ \sum_i \mathbb{E}_{x_i}\Big[R_i + \sum_z \tfrac{m_z}{2}(d_{iz} - \bar{d}_z)q_{iz}\Big] : \; x \in \mathcal{X},\; \bar{Q}(x) = Q \Big\}, \tag{4}$$

the carriers' *aggregate value function*, and

$$F(Q) = G(Q) + \sum_z \Big[-\alpha_z Q_z + \tfrac{m_z}{2} Q_z^2 + \tfrac{m_z}{2}\bar{d}_z Q_z \Big] + \iota_{\mathcal{Q}}(Q), \qquad g(Q) = \sum_z \Big[\tfrac{\kappa_z}{2} Q_z^2 - \tfrac{m_z}{2}\bar{d}_z Q_z \Big]. \tag{5}$$

For $v \in \mathbb{R}^Z$ and a positive vector w write $\|v\|_w^2 = \sum_z w_z v_z^2$ and $\langle v, v' \rangle_w = \sum_z w_z v_z v_z'$.

Lemma 2. [PROVED] *G is convex and piecewise linear on $\mathcal{Q}$. F is proper, closed, and strongly convex with modulus 1 in the norm $\|\cdot\|_m$: $F - \frac{1}{2}\|\cdot\|_m^2$ is convex. Moreover*

$$\min_{x \in \mathcal{X}} \overline{\Phi}_u(x) = \min_Q \; F(Q) + \langle u, Q \rangle, \qquad \min_{x \in \mathcal{X}} \big[\overline{\mathrm{SC}}(x) + L_{\bar{d}}(x)\big] = \min_Q \; F(Q) + g(Q),$$

where $L_{\bar{d}}(x) = \sum_{i,z} \frac{m_z}{2}(d_{iz} - \bar{d}_z)\bar{q}_{iz}(x_i)$, and in both cases x is a minimiser on the left if and only if $\bar{Q}(x)$ is a minimiser on the right and x attains the minimum in (4) at $Q = \bar{Q}(x)$.

Proof. G is the optimal value of a linear program in x whose right-hand side Q enters through the equality constraint $\bar{Q}(x) = Q$; such a value function is convex and piecewise linear on the set of right-hand sides for which the program is feasible, which is $\mathcal{Q}$. The quadratic part of F is $\frac{1}{2}\|Q\|_m^2$, the rest is convex, so $F - \frac{1}{2}\|\cdot\|_m^2 = G + \text{linear} + \iota_{\mathcal{Q}}$ is convex; F is closed because G is continuous on the closed set $\mathcal{Q}$. For the two identities, substitute $\sum_i \mathbb{E}_{x_i} \frac{m_z}{2} d_{iz} q_{iz} = \sum_i \mathbb{E}_{x_i} \frac{m_z}{2}(d_{iz} - \bar{d}_z)q_{iz} + \frac{m_z}{2}\bar{d}_z \bar{Q}_z$ in $\overline{\Phi}_u$, and $\frac{m_z^S}{2} = \frac{m_z}{2} + \frac{\kappa_z}{2}$ in $\overline{\mathrm{SC}} + L_{\bar{d}}$; in both the objective is a function of x only through $\bar{Q}(x)$ plus the quantity minimised in (4), so minimising over x equals minimising over $Q \in \mathcal{Q}$ the value $G(Q)$ plus the known function of Q, and the minimisers correspond as stated. □

Theorem 3 (The equilibrium response is a proximal map)**.** [PROVED] *For every $u \in \mathbb{R}^Z$ the convexified equilibria under charge u share one aggregate*

$$Q(u) = \arg\min_Q \big\{ F(Q) + \langle u, Q \rangle \big\} = \nabla F^*(-u),$$

where F^* is the convex conjugate of F, which is finite and differentiable on $\mathbb{R}^Z$. The response map is a contraction from the dual to the primal metric:

$$\|Q(u_1) - Q(u_2)\|_m \le \|u_1 - u_2\|_{m^{-1}} \qquad \text{for all } u_1, u_2. \tag{6}$$

Equivalently, $Q(u) = \text{prox}$ of G + linear in the m-metric: $Q(u) = \arg\min_Q G(Q) + \frac{1}{2}\|Q - m^{-1} \odot (\alpha - \frac{m}{2}\bar{d} - u)\|_m^2$.

Proof. By Lemma 2, $F + \langle u, \cdot \rangle$ is strongly convex on the compact set $\mathcal{Q}$, so it has a unique minimiser $Q(u)$, and every convexified equilibrium x has $\bar{Q}(x) = Q(u)$. For a closed proper strongly convex F the conjugate $F^*(y) = \sup_Q \{\langle y, Q \rangle - F(Q)\}$ is finite (the domain of F is bounded) and differentiable with $\nabla F^*(y) = \arg\max_Q \{\langle y, Q \rangle - F(Q)\}$ (Rockafellar, Theorem 26.3); with $y = -u$ this is $Q(u)$. For (6), the optimality conditions read $-u_k \in \partial F(Q_k)$, $k = 1, 2$. Strong convexity with modulus 1 in $\|\cdot\|_m$ gives, for all subgradients, $\langle (-u_1) - (-u_2), Q_1 - Q_2 \rangle \ge \|Q_1 - Q_2\|_m^2$ (apply the monotonicity of $\partial(F - \frac{1}{2}\|\cdot\|_m^2)$ and $\partial \frac{1}{2}\|\cdot\|_m^2 = m \odot \cdot$). By the Cauchy–Schwarz inequality in the m-inner product, $\langle u_2 - u_1, Q_1 - Q_2 \rangle = \langle m^{-1} \odot (u_2 - u_1), Q_1 - Q_2 \rangle_m \le \|u_2 - u_1\|_{m^{-1}} \|Q_1 - Q_2\|_m$, and (6) follows. The prox form is (5) with the square completed. □

Remark 4 (Link to ZMC). In the zone-market-clearing view (`theory/new_algorithm_A`) the equilibrium is a price λ at which the carriers' pricing answers clear every zone. Theorem 3 is the same object seen from the city: the city's instrument u is a shift of the zone price, the response $Q(u)$ is the gradient of the concave value function $u \mapsto -\min_x \overline{\Phi}_u(x)$, and the fixed-point equation of Theorem 5 is a supply line $u = \kappa Q - \frac{m}{2}\bar{d}$ against the response curve. The bracketed secant of ZMC could therefore be used by the leader; we use the gradient structure instead, because it gives the instance-free rate below.

Theorem 5 (Fixed point and dual problem)**.** [PROVED] *Let* $D(u) = F^*(-u) + g^*(u)$ *with* $g^*(u) = \sum_z \frac{(u_z + \frac{m_z}{2}\bar{d}_z)^2}{2\kappa_z}$. *Then*

(a) *D is finite, differentiable and strongly convex, with* $\nabla D(u) = -Q(u) + \kappa^{-1} \odot \big(u + \frac{m}{2}\bar{d}\big)$*: the gradient of the leader's dual objective is observable from the aggregate alone.*

(b) $\min_Q\{F(Q) + g(Q)\} = -\min_u D(u)$*, and* u^* *minimises* D *if and only if it is the unique fixed point of the tangent map* $T(u) = \kappa \odot Q(u) - \frac{m}{2} \odot \bar{d}$*. Its response* $Q^\infty = Q(u^*)$ *is the minimiser of* $F + g$*, i.e. of* $\overline{\mathrm{SC}} + L_{\bar{d}}$ *over* $\mathcal{X}$ *(Lemma 2).*

(c) *Welfare of the fixed point: every convexified equilibrium* x^∞ *under* u^* *satisfies* $\overline{\mathrm{SC}}(x^\infty) \le \min_x \overline{\mathrm{SC}}(x) + E_{\bar{d}}$*; with symmetric lots within zones (*$d_{iz} = \bar{d}_z$ *for all* i *with* $z \in C_i$*) it is a convexified social optimum.*

Proof. (a) $F^*(-\cdot)$ is finite and differentiable by Theorem 3 with gradient $-Q(u)$; g^* is the conjugate of the quadratic g, $g^*(u) = \sup_Q\{\langle u, Q\rangle - g(Q)\} = \sum_z (u_z + \frac{m_z}{2}\bar{d}_z)^2/(2\kappa_z)$ (maximise coordinatewise), with gradient $\kappa^{-1} \odot (u + \frac{m}{2}\bar{d})$ and Hessian $\mathrm{diag}(1/\kappa)$, so D is strongly convex. (b) Fenchel duality (Rockafellar, Corollary 31.2.1): $\inf_Q\{F(Q) + g(Q)\} = \sup_y\{-F^*(y) - g^*(-y)\}$ holds with attainment on the right because g is finite and continuous everywhere; with $y = -u$ the right side is $-\inf_u D(u)$. D is differentiable and strongly convex, so its minimiser u^* is unique and characterised by $\nabla D(u^*) = 0$, i.e. $u^* = \kappa \odot Q(u^*) - \frac{m}{2} \odot \bar{d} = T(u^*)$. Any fixed point of T is a zero of ∇D, hence the minimiser. At the saddle point the primal minimiser is $Q^\infty = \nabla F^*(-u^*) = Q(u^*)$ (Rockafellar, Theorem 31.3), and it minimises $F + g$, which is $\min\{\overline{\mathrm{SC}} + L_{\bar{d}}\}$ by Lemma 2. (c) Let x^* minimise $\overline{\mathrm{SC}}$. By (b), $\overline{\mathrm{SC}}(x^\infty) + L_{\bar{d}}(x^\infty) \le \overline{\mathrm{SC}}(x^*) + L_{\bar{d}}(x^*)$, so $\overline{\mathrm{SC}}(x^\infty) - \overline{\mathrm{SC}}(x^*) \le L_{\bar{d}}(x^*) - L_{\bar{d}}(x^\infty) \le \sum_{i,z} \frac{m_z}{2}|d_{iz} - \bar{d}_z|\,|\bar{q}^*_{iz} - \bar{q}^\infty_{iz}| \le E_{\bar{d}}$ because $\bar{q}_{iz} \in [0, d_{iz}]$. With symmetric lots $L_{\bar{d}} \equiv 0$. □

Theorem 6 (Instance-free contraction rates)**.** [PROVED] *Let* $\rho = \max_z \kappa_z/m_z = \max_z g_z r^2/(b_z + g_z r^2) < 1$ *over the zones with a lot, and let* u^* *be the fixed point of Theorem 5. In the convexified game, the rule* (3) *with a common* θ *satisfies:*

(a) (plain, $\theta = 1$) $\|u^{t+1} - u^*\|_{\kappa^{-1}} \le \rho\,\|u^t - u^*\|_{\kappa^{-1}}$ *for every* t *and every starting charge;*

(b) (damped, $0 < \theta \le \frac{2}{2+\rho}$) $\|u^{t+1} - u^*\|_{\kappa^{-1}} \le \sqrt{1 - \frac{2\theta(1+\rho)}{2+\rho}}\;\|u^t - u^*\|_{\kappa^{-1}}$*; at* $\theta = \frac{2}{2+\rho}$ *the ratio is* $\frac{\rho}{2+\rho} < \frac{1}{3}$.

The ratios depend on the instance only through ρ*: not on* Z*,* N*, the number of customers, the lots, the routing costs or the p-boxes. After* t *rounds the charge is within* $\rho^t\|u^0 - u^*\|_{\kappa^{-1}}$ *of the fixed point, and the equilibrium aggregate within* $\sqrt{\rho}\,\rho^t\|u^0 - u^*\|_{\kappa^{-1}}$ *of* Q^∞ *in the* $\|\cdot\|_m$ *norm.*

Proof. Let $K = \mathrm{diag}(\kappa)$ and consider $\mathbb{R}^Z$ with the inner product $\langle\cdot,\cdot\rangle_{\kappa^{-1}}$. The gradient of D with respect to this inner product is $K\nabla D(u) = u + \frac{m}{2}\bar{d} - \kappa \odot Q(u)$, so the rule (3) is $u^{t+1} = u^t - \theta\, K\nabla D(u^t)$: gradient descent on D with step θ in the κ^{-1}-geometry.

Two constants. Since $\kappa_z \le \rho m_z$ and $1/m_z \le \rho/\kappa_z$ for every active zone, (6) gives

$$\|Q(u_1) - Q(u_2)\|_\kappa \le \sqrt{\rho}\,\|Q(u_1) - Q(u_2)\|_m \le \sqrt{\rho}\,\|u_1 - u_2\|_{m^{-1}} \le \rho\,\|u_1 - u_2\|_{\kappa^{-1}}.$$

Hence $u \mapsto \kappa \odot Q(u)$ is ρ-Lipschitz from $\|\cdot\|_{\kappa^{-1}}$ to itself (note $\|\kappa \odot v\|_{\kappa^{-1}} = \|v\|_\kappa$), $K\nabla D$ is $(1+\rho)$-Lipschitz, and D is strongly convex with modulus 1 in this geometry (g^* contributes the identity, $F^*(-\cdot)$ is convex).

(a) $T(u) = \kappa \odot Q(u) - \frac{m}{2}\bar{d}$ is ρ-Lipschitz in $\|\cdot\|_{\kappa^{-1}}$ and $u^* = T(u^*)$, so $\|u^{t+1} - u^*\|_{\kappa^{-1}} = \|T(u^t) - T(u^*)\|_{\kappa^{-1}} \le \rho\|u^t - u^*\|_{\kappa^{-1}}$.

(b) For a function that is μ-strongly convex with L-Lipschitz gradient in a Euclidean geometry, gradient descent with step $\theta \le 2/(\mu+L)$ satisfies $\|u^{t+1} - u^*\|^2 \le \big(1 - \frac{2\theta\mu L}{\mu+L}\big)\|u^t - u^*\|^2$ (Nesterov, *Introductory Lectures*, Theorem 2.1.15; the proof uses only the inequality $\langle \nabla D(u) - \nabla D(u^*), u - u^*\rangle \ge \frac{\mu L}{\mu+L}\|u - u^*\|^2 + \frac{1}{\mu+L}\|\nabla D(u) - \nabla D(u^*)\|^2$, valid in any inner-product space). With $\mu = 1$, $L = 1+\rho$ the factor is $1 - 2\theta(1+\rho)/(2+\rho)$; at $\theta = 2/(2+\rho)$ it equals $1 - 4(1+\rho)/(2+\rho)^2 = \rho^2/(2+\rho)^2$.

The last claim follows from (a) and $\|Q(u) - Q(u^*)\|_m \le \|u - u^*\|_{m^{-1}} \le \sqrt{\rho}\|u - u^*\|_{\kappa^{-1}}$. □

Remark 7. The plain iteration is the classical trial-and-error implementation of marginal-cost pricing (Yang, Meng and Lee 2004) transferred to a Cournot market with lumpy lots; its proof of convergence there uses the method of successive averages and gives no rate. The rate ρ comes from the Cournot slope b_z: without the demand side ($b = 0$) one has $\rho = 1$ and no contraction, which is why the nonatomic routing literature needs averaging. In the city games ρ is 0.03–0.4 (`results.md`), so two or three rounds bring the charge within a few percent of its fixed point.

3 The pure, lumpy game

In the pure game the carriers settle, after a charge is posted, in some pure equilibrium $\sigma^t \in \mathrm{NE}(u^t)$ (in the experiments: best-response dynamics from the previous profile, a descent of Φ_{u^t}), and the city observes $Q^t = Q(\sigma^t)$. Pure responses are set-valued and piecewise constant, so a fixed point need not exist. Two things can be said exactly.

Theorem 8 (Fixed points are unilateral social optima up to lot asymmetry)**.** [PROVED] *Let $\sigma \in \mathrm{NE}(u)$ with $u = \kappa \odot Q(\sigma) - \frac{m}{2} \odot \bar{d} + v$. Then for every carrier i and every plan $p \in \mathcal{P}_i$, with $\sigma' = (p, \sigma_{-i})$ and $\Delta Q = Q(\sigma') - Q(\sigma)$,*

$$\mathrm{SC}(\sigma') \;\ge\; \mathrm{SC}(\sigma) + \big[L_{\bar{d}}(\sigma) - L_{\bar{d}}(\sigma')\big] + \sum_z \tfrac{\kappa_z}{2}\Delta Q_z^2 - \langle v, \Delta Q\rangle.$$

At an exact fixed point ($v = 0$) with symmetric lots, no carrier can lower the social cost unilaterally; in general the possible unilateral gain is at most $L_{\bar{d}}(\sigma') - L_{\bar{d}}(\sigma) \le \sum_z \frac{m_z}{2}|d_{iz} - \bar{d}_z|d_{iz}$, the asymmetry of the deviating carrier's own lots.

Proof. Apply Lemma 1 with $Q^\circ = Q(\sigma)$. Since σ is an equilibrium, $\Phi_u(\sigma) \le \Phi_u(\sigma')$ (Monderer–Shapley: a unilateral deviation changes Φ_u by the deviator's cost change). The $(Q - Q^\circ)^2$ term vanishes at σ and equals $\sum_z \frac{\kappa_z}{2}\Delta Q_z^2$ at σ'; rearranging (2) for σ and σ' gives the inequality. The last bound uses $|q_{iz}(p) - q_{iz}(\sigma_i)| \le d_{iz}$. □

Theorem 9 (The pure iteration stays near the convexified fixed point)**.** [PROVED] *Run the plain rule on the pure game and let $e^t = Q^t - Q(u^t)$ be the gap between the observed pure aggregate and the convexified response. Then*

$$\|u^{t+1} - u^*\|_{\kappa^{-1}} \le \rho\,\|u^t - u^*\|_{\kappa^{-1}} + \|e^t\|_\kappa, \qquad \limsup_t \|u^t - u^*\|_{\kappa^{-1}} \le \frac{\sup_t \|e^t\|_\kappa}{1-\rho}.$$

If the carriers settle in a potential-minimising equilibrium, $\sigma^t \in \arg\min_\sigma \Phi_{u^t}(\sigma)$, then $\|e^t\|_m \le \sqrt{2\eta_t}$ with $\eta_t = \Phi_{u^t}(\sigma^t) - \min_{x\in\mathcal{X}} \overline{\Phi}_{u^t}(x) \ge 0$ the rounding gap of the potential, and $\|e^t\|_\kappa \le \sqrt{2\rho\,\eta_t}$.

Proof. $u^{t+1} = \kappa \odot Q^t - \frac{m}{2}\bar{d} = T(u^t) + \kappa \odot e^t$ and $u^* = T(u^*)$, so the first inequality follows from Theorem 6(a) and $\|\kappa \odot e\|_{\kappa^{-1}} = \|e\|_\kappa$; iterate it for the limsup. For the gap bound, let $h = F + \langle u^t, \cdot\rangle$, which is strongly convex with modulus 1 in $\|\cdot\|_m$ and minimised at $Q(u^t)$, so $h(Q) \ge h(Q(u^t)) + \frac{1}{2}\|Q - Q(u^t)\|_m^2$ for all $Q \in \mathcal{Q}$. The pure profile σ^t is a point of $\mathcal{X}$ with $\bar{Q} = Q^t$, so $\overline{\Phi}_{u^t}(\sigma^t) \ge h(Q^t)$ by (4), while $\min_x \overline{\Phi}_{u^t} = h(Q(u^t))$ by Lemma 2; and $\Phi_{u^t}(\sigma^t) = \overline{\Phi}_{u^t}(\sigma^t)$ at a pure profile. Hence $\eta_t \ge \frac{1}{2}\|e^t\|_m^2$, and $\|e^t\|_\kappa \le \sqrt{\rho}\|e^t\|_m$. □

η_t is observed, not predicted; the a priori bounds of `main.tex` (Theorem on D1 rounding, $\eta \le \frac{1}{2}\min\{N, d_L\}R_M^2$) apply to the potential minimiser only. For other pure equilibria no bound holds, and the experiments report what the carriers' dynamics select.

Theorem 10 (What aggregate observations identify)**.** [PROVED]

(a) The convexified response map $u \mapsto Q(u)$ on $\mathbb{R}^Z$ determines F, hence G, up to an additive constant. Two instances with the same (b, g, r) and the same response map have the same convexified social optimum aggregate and the same fixed point u^*.

(b) The pure social optimum is not determined by the convexified response map: there are two instances with identical F whose pure social optima differ in aggregate and in value (Example 11).

Consequently the fixed point of Theorem 5 (the minimiser of $\overline{\mathrm{SC}} + L_{\bar{d}}$) is the most that any scheme using the convexified response can target, and the plain rule reaches it at the rate of Theorem 6.

Proof. (a) By Theorem 3, $Q = Q(u)$ if and only if $-u \in \partial F(Q)$. F is polyhedral plus a quadratic on the polytope $\mathcal{Q}$, so $\partial F(Q) \ne \emptyset$ for every $Q \in \mathcal{Q}$ (Rockafellar, Theorem 23.10 applied to the polyhedral part, the quadratic part being differentiable). Therefore the graph of the response map, $\{(u, Q(u))\}$, is the graph of $-\partial F$ restricted to $\mathcal{Q}$, and $\mathcal{Q} = \{Q(u) : u \in \mathbb{R}^Z\}$. A closed proper convex function is determined by its subdifferential up to an additive constant (Rockafellar, Theorem 24.9). The convexified optimum minimises $F + g$ with g known, and u^* is the unique zero of ∇D, both unchanged by a constant shift of F. (b) Example 11. □

Example 11. [PROVED] One zone, $\kappa = 0$ (no congestion), slope $b = m$, intercept α, reference $\bar{d} = d$. Instance A: two carriers, each with the lot d and the plans "stay out" ($R = 0$) and "serve" ($R = r$). Instance B: one carrier with the lot $2d$ and the plans "stay out" ($R = 0$) and "serve" ($R = 2r - md^2$), with $r > md^2/2$. In both, $\mathcal{Q} = [0, 2d]$ and G is linear: $G_A(Q) = \frac{r}{d}Q + 0$ (the own term $\frac{m}{2}(d - \bar{d})q$ vanishes) and $G_B(Q) = \frac{2r - md^2}{2d}Q + \frac{m}{2}(2d - d)Q = \frac{r}{d}Q$. So $F_A = F_B$ and the response maps coincide. The pure social costs are $\mathrm{SC}_A(Q) = \frac{r}{d}Q - \alpha Q + \frac{m}{2}Q^2$ on $Q \in \{0, d, 2d\}$ and $\mathrm{SC}_B(Q) = (2r - md^2)\mathbf{1}[Q = 2d] - \alpha Q + \frac{m}{2}Q^2$ on $Q \in \{0, 2d\}$. Choose $\alpha = \frac{r}{d} + md$: then $Q = d$ is the unique minimiser of the continuous relaxation $F + g$ and the unique pure optimum of A, with value $\mathrm{SC}_A(d) = r - \alpha d + \frac{m}{2}d^2 = -\frac{m}{2}d^2$ (and $\mathrm{SC}_A(0) = \mathrm{SC}_A(2d) = 0$), while B has the unique pure optimum $Q = 2d$ with value $\mathrm{SC}_B(2d) = 2r - md^2 - 2\alpha d + 2md^2 = -md^2$. The optima differ in aggregate and in value although nothing observable in the convexified game differs.

4 Carriers with imprecise beliefs

Proposition 12 (Γ-maximin surcharge)**.** [PROVED] *Under Γ-maximin beliefs of width δ about the rivals' aggregate (`main.tex`, Lemma on mixtures), carrier i's cost gains the surcharge $\delta \sum_k M_k w_k a_{ik}(p) =$*

$\langle \hat{s}(\delta), q_i(p) \rangle$ *with* $\hat{s}_z(\delta) = \delta(b_z w_{z,Q} + g_z r\, w_{z,X})$*, linear in the deliveries. Hence the response of the* δ*-game to the charge* u *is the response of the precise game to* $u + \hat{s}(\delta)$*, the rule* (3) *needs no knowledge of* δ *and contracts at the rates of Theorem 6, and its fixed point* u^*_δ *is the unique solution of* $u = T(u + \hat{s}(\delta))$*; its response minimises* $\overline{\mathrm{SC}} + L_{\bar{d}} + \langle \hat{s}(\delta), \bar{Q} \rangle$*. If the city knows* $\hat{s}(\delta)$ *(it knows* w *and* δ*), posting* $u - \hat{s}(\delta)$ *instead of* u *reproduces the precise fixed point exactly, as in DIP's Theorem 12. Otherwise the loss in* $\overline{\mathrm{SC}} + L_{\bar{d}}$ *relative to the precise fixed point is at most* $\frac{1}{2}\|\hat{s}(\delta)\|^2_{m^{-1}}$*.*

Proof. The surcharge is a per-delivery charge, so the δ-game under u is the precise game under $u + \hat{s}$ (DIP, Theorem 12). The map $u \mapsto T(u + \hat{s})$ is ρ-Lipschitz, so the plain rule contracts as before; a fixed point u^*_δ makes $w = u^*_\delta + \hat{s}$ a zero of $\nabla D_{\hat{s}}$ with $D_{\hat{s}}(w) = F^*(-w) + g^*(w - \hat{s})$, the dual of $\min_Q F(Q) + g(Q) + \langle \hat{s}, Q \rangle$, which is unique by strong convexity. For the loss, let Q_0 and Q_δ be the minimisers of $H = F + g$ and of $H + \langle \hat{s}, \cdot \rangle$; both functions are strongly convex with modulus 1 in $\|\cdot\|_m$. Strong convexity of $H + \langle \hat{s}, \cdot \rangle$ at its minimiser Q_δ gives $H(Q_0) + \langle \hat{s}, Q_0 \rangle \geq H(Q_\delta) + \langle \hat{s}, Q_\delta \rangle + \frac{1}{2}\|Q_0 - Q_\delta\|^2_m$, i.e. $H(Q_\delta) - H(Q_0) \leq \langle \hat{s}, Q_0 - Q_\delta \rangle - \frac{1}{2}\|Q_0 - Q_\delta\|^2_m \leq \|\hat{s}\|_{m^{-1}}\|Q_0 - Q_\delta\|_m - \frac{1}{2}\|Q_0 - Q_\delta\|^2_m \leq \frac{1}{2}\|\hat{s}\|^2_{m^{-1}}$. By Lemma 2, H is $\overline{\mathrm{SC}} + L_{\bar{d}}$ minimised over the profiles with the given aggregate. □

Proposition 13 (E-admissible absorption of the leader's error)**.** [PROVED] *Let* σ^* *be a pure Nash equilibrium under the charge* u^* *(precise beliefs). Let the carriers hold credal boxes of width* δ *around the realised rival aggregate (`main.tex`, Assumption on credal sets, weights* w*) and let* $\hat{s}(\delta)$ *be as in Proposition 12. Write* $\hat{w}_z = b_z w_{z,Q} + g_z r\, w_{z,X}$*, so that* $\hat{s}_z(\delta) = \delta \hat{w}_z$*. If a posted charge* u *satisfies, for every zone* z *and every carrier* i *with* $z \in C_i$*,*

$$|u_z - u^*_z| \leq \hat{s}_z(\delta) \quad \textit{and} \quad \Big(u_z \leq u^*_z \;\; \textit{or} \;\; Q_{-i,z}(\sigma^*) \geq (u_z - u^*_z)\,\frac{\max\{w_{z,Q},\, w_{z,X}/r\}}{\hat{w}_z}\Big),$$

then $\sigma^* \in \mathrm{ICNE}_E(\delta)$ *under* u*: each carrier's plan is optimal for some belief in its box. In particular the plain rule, started anywhere in the convexified game, places* σ^* *in* $\mathrm{ICNE}_E(\delta)$ *as soon as* $\rho^t\|u^0 - u^*\|_{\kappa^{-1}} \leq \min_z \hat{s}_z(\delta)/\sqrt{\kappa_z}$*, i.e. after* $\lceil \log(\|u^0 - u^*\|_{\kappa^{-1}} \max_z \sqrt{\kappa_z}/\hat{s}_z(\delta))/\log(1/\rho) \rceil$ *rounds.*

Proof. Carrier i's cost of a plan p at the charge u and the belief (μ_Q, μ_X) about the rivals' zone deliveries and hours is $R_i(p) + \sum_z q_{iz}(p)\big(-\alpha_z + m_z d_{iz} + b_z \mu_{Q,z} + g_z r \mu_{X,z} + u_z\big)$. Write $\Delta_z = u_z - u^*_z$ and pick the belief $\mu_{Q,z} = Q_{-i,z} - \Delta_z w_{z,Q}/\hat{w}_z$, $\mu_{X,z} = X_{-i,z} - \Delta_z w_{z,X}/\hat{w}_z$. Then $b_z\mu_{Q,z} + g_z r\mu_{X,z} + u_z = b_z Q_{-i,z} + g_z r X_{-i,z} - \Delta_z(b_z w_{z,Q} + g_z r w_{z,X})/\hat{w}_z + u_z = b_z Q_{-i,z} + g_z r X_{-i,z} + u^*_z$, so the cost of every plan at this belief and charge u equals its cost at the realised aggregate and charge u^*, where σ^*_i is optimal. The belief lies in the box because $|\mu_{Q,z} - Q_{-i,z}| = |\Delta_z| w_{z,Q}/\hat{w}_z \leq \delta w_{z,Q}$ and $|\mu_{X,z} - X_{-i,z}| = |\Delta_z| w_{z,X}/\hat{w}_z \leq \delta w_{z,X}$ exactly when $|\Delta_z| \leq \delta \hat{w}_z = \hat{s}_z(\delta)$. It is nonnegative when $\Delta_z \leq 0$; when $\Delta_z > 0$ it needs $Q_{-i,z} \geq \Delta_z w_{z,Q}/\hat{w}_z$ and $X_{-i,z} = rQ_{-i,z} \geq \Delta_z w_{z,X}/\hat{w}_z$, both implied by the stated condition. Hence σ^*_i is E-admissible for every i (`main.tex`, reduction of the credal set to its mean box). The round count follows from Theorem 6(a) and $|u_z - u^*_z| \leq \sqrt{\kappa_z}\|u - u^*\|_{\kappa^{-1}}$. □

5 The limit of re-tangenting: the realised-aggregate tariff

The flat rule charges every delivery in z the externality $\kappa_z Q_z$ at the observed aggregate minus a half-lot subsidy estimated from n_z. The exact externality of carrier i's lot on the others is $\kappa_z d_{iz} Q_{-i,z}$ and its market-power wedge is $\frac{b_z}{2} d^2_{iz}$; both are functions of its own deliveries and of the zone total, so an anonymous nonlinear tariff can charge them exactly.

Theorem 14 (Marginal tariff)**.** [PROVED] *(the mechanism is classical: Wolpert–Tumer 1999, Sandholm 2002, Marden–Wierman 2013; the lumpy form and its consequences for the city game are stated here) Let every carrier pay, in every zone,* $T_z(q_{iz}, Q_z) = \kappa_z q_{iz}(Q_z - q_{iz}) - \frac{b_z}{2} q^2_{iz}$*. Then:*

(a) for all i, p and σ_{-i}: $J_i^T(p;\sigma_{-i}) = \mathrm{SC}(p,\sigma_{-i}) - \mathrm{SC}(\emptyset_i,\sigma_{-i})$, so SC *is an exact potential of the tariff game;*

(b) the pure equilibria are exactly the profiles that no single carrier can improve socially; every social optimum is a pure equilibrium (price of stability 1*); best-response dynamics from any profile never increase* SC *and stop in finitely many steps at such an equilibrium;*

(c) given the rivals, the tariff is linear in the carrier's own service indicators, so a best response is one call of the routing oracle with the zone prices shifted by $\kappa_z Q_{-i,z} - \frac{b_z}{2} d_{iz}$;

(d) the tariff equals the flat fixed-point charge with the carrier's own lot in place of $\bar{d}_z$ minus the own-congestion rebate $\frac{\kappa_z}{2} d_{iz}^2$: $T_z(d_{iz}, Q_z) = d_{iz}\big(\kappa_z Q_z - \frac{m_z}{2} d_{iz}\big) - \frac{\kappa_z}{2} d_{iz}^2$. The city's net revenue is $\sum_z [\kappa_z Q_z^2 - (\kappa_z + \frac{b_z}{2}) \sum_i q_{iz}^2]$.

Proof. (a) $J_i^T(p;\sigma_{-i}) = R_i(p) + \sum_z q_{iz}\big(-\alpha_z + m_z d_{iz} + m_z Q_{-i,z} + \kappa_z Q_{-i,z} - \frac{b_z}{2} d_{iz}\big) = R_i(p) + \sum_z q_{iz}\big(-\alpha_z + \frac{m_z^S}{2} d_{iz} + m_z^S Q_{-i,z}\big)$, using $q_{iz}^2 = d_{iz} q_{iz}$, $m_z - \frac{b_z}{2} = \frac{m_z^S}{2}$ and $m_z + \kappa_z = m_z^S$. On the other side, $\mathrm{SC}(p,\sigma_{-i}) - \mathrm{SC}(\emptyset_i,\sigma_{-i}) = R_i(p) + \sum_z \big[-\alpha_z q_{iz} + \frac{m_z^S}{2}\big((q_{iz} + Q_{-i,z})^2 - Q_{-i,z}^2\big)\big] = R_i(p) + \sum_z q_{iz}\big(-\alpha_z + \frac{m_z^S}{2} d_{iz} + m_z^S Q_{-i,z}\big)$. The two agree, and the right side differs from $\mathrm{SC}(p,\sigma_{-i})$ by a term independent of p, which is the definition of an exact potential. (b) Monderer–Shapley for the potential SC on a finite profile set. (c) Immediate from the first expression in (a). (d) $\kappa_z d(Q_z - d) - \frac{b_z}{2} d^2 = d(\kappa_z Q_z - \frac{m_z}{2} d) - \frac{\kappa_z}{2} d^2$ since $\kappa_z + \frac{b_z}{2} = \frac{m_z}{2} + \frac{\kappa_z}{2}$; summing T_z over carriers gives the revenue. □

Remark 15. Under the tariff the whole machinery of DIP (social target, cutting planes, fallback) is unnecessary: the target is an equilibrium by (b), for any lot asymmetry. What remains is equilibrium selection, which is now a pure optimisation question about the potential SC: the carriers' best-response dynamics from the status quo reach a unilateral social optimum that is never worse than the status quo (observation-only), and the potential-minimising solver of E3 (ZMC on the social-cost potential) reaches the same profile DIP would implement, with no implementation step. The tariff needs the city to bill each carrier on its own zone deliveries, which any per-delivery charge needs too, and to know the zone total, which it observes.

6 Estimating the mean lot from the observation

Lemma 16. [PROVED] *(a) If lots are symmetric within zones, $Q_z/n_z = \bar{d}_z$ exactly in every served zone. (b) If between two observations n_z changes by exactly one, $|Q_z^t - Q_z^{t-1}|$ is the lot of the carrier that entered or left; if $n_z = 1$, Q_z is the lot of the serving carrier. (c) For a zone never served the observation carries no information about its lots; the rule uses the mean observed lot of the city, and Theorem 5(c) bounds the cost of the error by $E_{\bar{d}}$ with $\bar{d}$ the vector actually used.*

Proof. (a) $Q_z = n_z \bar{d}_z$. (b) All other carriers' lots in z are unchanged, so the difference of the sums is the changed term. (c) Nothing distinguishes two instances that differ only in the lots of a never-served zone. The bound is Theorem 5(c), whose proof holds for any reference vector. □

7 Prior art and what is new

Classical. Marginal-cost pricing by trial and error on observed flows: Yang, Meng and Lee (TR-B 2004; method of successive averages, nonatomic networks, no rate); Han and Yang (2008), Ye, Yang

and Tan (TR-B 2015, disequilibrium flows), Xu, Meng and Huang (TR-C 2016, global convergence), Zhou et al. (2015, Lagrangian-dual view with capacity constraints). Adaptive incentive design with two-timescale learning: Maheshwari, Kulkarni, Wu and Sastry (2022, atomic Cournot and quadratic aggregative games with continuous strategies; fixed points are socially optimal; convergence by stochastic approximation, no rate). Learning optimal taxes from equilibrium feedback in nonatomic congestion games: Cui, Fazel and Du (NeurIPS 2024). Linear convergence of projection-descent methods for Stackelberg aggregative games under strong monotonicity, with continuous follower strategies and a leader who knows the followers' responses: Li, Li, Li and Chen (IEEE TAC 2025). Marginal-contribution tariffs that make welfare the potential: Wolpert and Tumer (1999), Sandholm (REStud 2002; JET 2007, evolutionary implementation of Pigouvian pricing, nonatomic), Marden and Wierman (2013). Pigou–Cournot corrective taxes: Buchanan (1969), Barnett (1980). Convex duality, conjugates and proximal maps: Rockafellar (1970), Moreau (1965).

New here. (i) The identification of the convexified equilibrium-aggregate map of the lumpy Cournot–congestion routing game with a proximal map, $Q(u) = \nabla F^*(-u)$, and of the leader's problem with the strongly convex dual D whose gradient is the observed aggregate (Theorems 3–5); (ii) the instance-free contraction ratios ρ and $\rho/(2+\rho)$ for the plain and damped Pigou–Cournot iteration, dimension-free and routing-free (Theorem 6); (iii) the lumpy fixed-point property (Theorem 8), the neighbourhood bound through the rounding gap (Theorem 9) and the identification limit with its example (Theorem 10); (iv) the absorption of the leader's search error by the carriers' credal width (Proposition 13) and the surcharge shift of the fixed point (Proposition 12); (v) the lumpy form of the marginal tariff and the observation that it removes DIP's implementation step entirely (Theorem 14). The central mechanism, charge the externality at the observed aggregate, is not new; its theory in this setting is.

Not claimed. No guarantee on which pure equilibrium the carriers select under a posted charge; no bound for pure equilibria that are not potential minimisers (Theorem 9 is conditional); no result for D2; no pessimistic (worst-equilibrium) guarantee beyond Theorem 14(b), which only says that every tariff equilibrium is a unilateral social optimum.